The Shape of a Knowledge Boundary: A Transformer Trained Only on Pre-1939 Text

Andrew An
1010864165
`https://github.com/Shinoaki798/1939` I used Claude Fable 5.1 to do the proposal.

Introduction

A language model's knowledge cutoff is usually described as a wall. I suspect it is a **membrane with structure**: some post-cutoff concepts are unreachable, others are assembled from parts the model already has and are half predictable. To characterise it I will train a **decoder-only Transformer language model** from random initialisation on text ending before the Second World War — American and German press, the German in the original, because the country that started the war kept its own record — and ask three questions. **RQ1 (permeability):** which classes of post-1939 concepts can a pre-1939 model reach? **RQ2 (perspective):** what does reading the German press change — measured against an otherwise identical twin trained on English only? **RQ3 (foresight):** how accurately does a model that has read everything up to June 1939 predict what happened next — the war, a third Roosevelt term, nuclear weapons — against what informed people believed at the time? Deep learning is the instrument: these quantities exist only in a trained model's next-token distribution. Training ends **30 June 1939**; July–August 1939 is an **embargo**, excluded from training and scoring.

Illustration

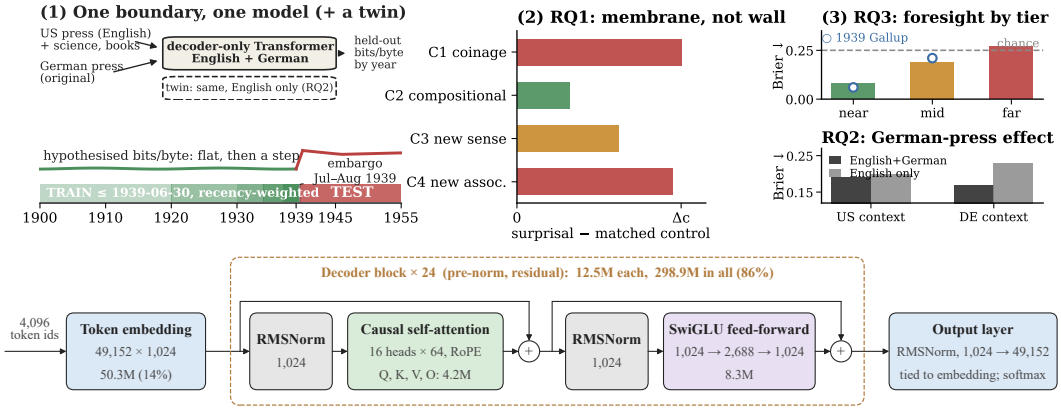


Figure 1: Top: one boundary, one model and its English-only twin (shapes hypothesised, not results). Bottom: the Transformer of both runs, each layer with its size.

Background & Related Work

Language models degrade on text written after their training window [16]; temporal conditioning improves future-fact calibration [3]; Time Machine GPT [4] trains point-in-time models. Recent from-scratch models with 19th- and 20th-century cutoffs, including 1939 [9, 10, 20], aim to *produce* such models; this project measures the boundary. SemEval-2020 Task 1 [21], HistBERT [19] and 19th-century models [12] measure how *existing* words drift with masked encoders; RQ1 asks which *not-yet-existing* concepts a fixed earlier vocabulary can reach, which needs a causal model with a hard boundary. Gallup's 1939 polls [7] give RQ3 a hindsight-free human reference.

Data Processing

Sources. All text is pre-cutoff and in the original language. *English:* American Stories [1], article-level text from Chronicling America, is the backbone and sole source of scored sets; training-only additions: Chronicling America batch OCR for the 1930s [18], the Congressional Record [8], the Federal Register [22] and case law [11] ($\leq$10%), and public-domain books [17] ($\leq$12%). *German:* newspapers from Europeana [6] and the Deutsches Zeitungsportal [2], deduplicated by title and date, plus the NSDAP daily *Völkischer Beobachter* (1925, 1930) [13]; German is $\leq$25% of seen tokens per period. *Science* ($\leq$10%, no recency weighting): JSTOR Early Journal Content [15], Internet Archive scans of *Annalen der Physik*, *Scientific American*, *Nature* and the 1911 Britannica [14], and the JFM mathematics reviews [5]. **Recency weighting**, as the model must live in 1939: 1934–39 is seen twice, the rest of the 1930s once; the 1920s fill the remainder ($\leq$35%); pre-1920 $\leq$8%; within a period, weight $\propto \exp(-(1939 - \text{year})/5)$. **Splits** by publication date, never shuffled: a fixed 2% of every year 1900–1955, per language, is held out; TRAIN = the rest $\leq$1939-06-30; VAL = a disjoint 2% $\leq$ cutoff; TEST-A/B/C = held-out 1930–38 / 1939-09–45 / 1946–55. **Cleaning**: (1) language filter (stopword vote); (2) MinHash dedup of reprints (word 5-grams, Jaccard $\geq$0.8) before splitting; (3) OCR quality per language: period-lexicon hit rate $\geq$0.75 and word-like share $\geq$0.70, $\geq$50 tokens; (4) C1 coinages are screened and counted; pre-cutoff hits (surnames, OCR noise) are period text and stay. **Format:** 48k BPE trained on native pre-cutoff English and German; 4096-token sequences; all numbers are bits-per-byte. **RQ1 probe set:** $\sim$200 post-cutoff terms in four classes (C1 coinage *radar*; C2 compositional *atomic bomb*; C3 new sense *blitz*; C4 new association *Hiroshima*), each with a frequency-matched control in $\geq$5 contexts, frozen in advance. **RQ3 propositions:** $\geq$120 binary propositions resolved after 1939-09-01, in three tiers by what was knowable in June 1939 — *near* (war in Europe this year), *mid* (a third Roosevelt term, a chain reaction), *far* (a fission weapon used) — each with same-topic US (English) and German contexts from June and late August 1939 (embargo as context only), three wordings each, English continuations, and the Gallup figure where polled.

Architecture

A **decoder-only Transformer** (Figure 1, bottom), a next-token predictor written from scratch and trained from random initialisation; decoder-only because every measurement is the probability of unseen text, which a masked encoder does not give. **Two runs, one architecture:** the main model (English+German) and, for RQ2 only, a **twin** identical in size, tokenizer, token budget and schedule, its German slice replaced by English of the same periods. **Size:** 349M parameters on 12B seen tokens, no text seen more than twice; $\approx 3 \times 10^{19}$ FLOPs per run: $\sim$130 h on the RTX 5080 or $\sim$22 h on a rented H100. **Measurement.** RQ1: for term class c, $\Delta_c = \mathbb{E}_{t \in \mathcal{T}_c}[s(t)] - \mathbb{E}_{t \in \mathcal{C}_c}[s(t)]$, s the surprisal in fixed contexts, $\mathcal{C}_c$ matched controls, plus a C4 forced choice (naval station vs. attack) that doubles as a contamination check. RQ3: forced choice between two continuations, the normalised log-likelihood ratio as probability, Brier per tier, perspective and window against chance, the n-gram floor and Gallup. RQ2: the same batteries on the twin; the paired difference is the German-press effect. All with bootstrap 95% CIs. **Success:** Δ_{C2} significantly smallest and a step at 1939 (RQ1); mid-tier Brier better than chance and not worse than Gallup (RQ3); a significant paired difference on the English batteries — US-context propositions, Δ_c, per-year bpb (RQ2).

Baseline Model

With the same split, tokenizer and budget: a **vanilla RNN**, an **LSTM** and a **GRU** language model (2 layers, $\sim$50M parameters, next-token objective), which justify the Transformer; a **modified Kneser–Ney 5-gram** (KenLM defaults) is the non-neural floor. **Plan** (individual): corpus frozen 20 Oct; main Transformer 22–31 Oct; twin and baselines 1–12 Nov.

Ethical Considerations

Pre-1939 newspapers carry the racial, ethnic and gender language of their time; the model reproduces it, documented, not sanitised. German press after 1933 was state-controlled: the "German perspective" is what Germans were permitted to read, not a verdict on a nation. In collection, OCR noise concentrates in older and Fraktur text. Sources are public domain or openly licensed; the model is released non-commercially, not for querying the private people the papers name.

References

[1] M. Dell, J. Carlson, T. Bryan, E. Silcock, A. Arora, Z. Shen, L. D'Amico-Wong, Q. Le, P. Querubin, and L. Heldring. American stories: A large-scale structured text dataset of historical U.S. newspapers. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2023.

[2] Deutsche Digitale Bibliothek. Deutsches Zeitungsportal. `https://www.deutsche-digitale-bibliothek.de/newspaper`, 2024.

[3] B. Dhingra, J. R. Cole, J. M. Eisenschlos, D. Gillick, J. Eisenstein, and W. W. Cohen. Time-aware language models as temporal knowledge bases. *Transactions of the Association for Computational Linguistics*, 10, 2022.

[4] F. Drinkall, E. Rahimikia, J. B. Pierrehumbert, and S. Zohren. Time machine GPT. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 2024.

[5] European Mathematical Society and FIZ Karlsruhe. Jahrbuch über die Fortschritte der Mathematik (1868–1942), via zbMATH Open. `https://zbmath.org/`, 2021. CC BY-SA 4.0.

[6] Europeana Foundation. Europeana Newspapers: Full-Text Collection. `https://www.europeana.eu/en/collections/topic/18-newspapers`, 2015. Hugging Face mirror: biglam/europeana_newspapers.

[7] G. H. Gallup. The Gallup Poll: Public Opinion 1935–1971. Random House, New York, 1972.

[8] M. Gentzkow, J. M. Shapiro, and M. Taddy. Congressional Record for the 43rd–114th Congresses: Parsed Speeches and Phrase Counts. Stanford Libraries, 2018.

[9] D. Göttlich, D. Loibner, G. Jiang, and H.-J. Voth. Ranke-4B: History LLMs with fixed knowledge cutoffs. `https://github.com/DGoettlich/history-llms`, 2025.

[10] H. Grigorian. TimeCapsuleLLM: A language model trained only on data from 1800–1875. `https://github.com/haykgrigo3/TimeCapsuleLLM`, 2025.

[11] Harvard Law School Library. Caselaw Access Project. `https://static.case.law/`, 2024.

[12] K. Hosseini, K. Beelen, G. Colavizza, and M. Coll Ardanuy. Neural language models for nineteenth-century English. *Journal of Open Humanities Data*, 7, 2021.

[13] Internet Archive. Völkischer Beobachter, 1925 and 1930 issues. `https://archive.org/details/pub_volkischer-beobachter`, 2024.

[14] Internet Archive. Scanned volumes with full-text OCR: Annalen der Physik, Scientific American, Nature, Proceedings of the National Academy of Sciences, Encyclopædia Britannica (11th edition) and further pre-1939 science journals. `https://archive.org/`, 2026.

[15] JSTOR. JSTOR Early Journal Content, data bundle (STEM journal titles used). `https://archive.org/details/jstor_ejc_data_bundle`, 2011.

[16] A. Lazaridou, A. Kuncoro, E. Gribovskaya, D. Agrawal, A. Liska, T. Terzi, M. Gimenez, C. de Masson d'Autume, T. Kocisky, S. Ruder, D. Yogatama, K. Cao, S. Young, and P. Blunsom. Mind the gap: Assessing temporal generalization in neural language models. In *Advances in Neural Information Processing Systems*, 2021.

[17] Library of Congress. Library of Congress Public Domain Books. `https://www.loc.gov/collections/selected-digitized-books/`, 2024. Hugging Face mirror: storytracer/LoC-PD-Books.

[18] Library of Congress. Chronicling America: Historic American Newspapers, batch OCR. `https://chroniclingamerica.loc.gov/data/ocr/`, 2026.

[19] W. Qiu and Y. Xu. HistBERT: A pre-trained language model for diachronic lexical semantic analysis. *arXiv preprint arXiv:2202.03612*, 2022.

[20] A. Radford, N. Levine, and D. Duvenaud. Introducing talkie: A 13B vintage language model from 1930. `https://talkie-lm.com/introducing-talkie`, 2026.

[21] D. Schlechtweg, B. McGillivray, S. Hengchen, H. Dubossarsky, and N. Tahmasebi. SemEval-2020 task 1: Unsupervised lexical semantic change detection. In *Proceedings of the 14th International Workshop on Semantic Evaluation*, 2020.

[22] U.S. Government Publishing Office. Federal Register, 1936–1939. `https://www.govinfo.gov/app/collection/fr`, 2024.